

Basic Principles of Medicine 2

Module: Digestion and Metabolism
Lecture No.39

Combined Physiology and Biochemistry
Clinical Case

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Required Reading

Handout on absorption

The case study is retrieved and modified from

Physiology Cases and Problems
4th edition, Wolters Kluwer, Lippincott Williams & Wilkins

Chapter 5 Gastrointestinal Physiology
Case 48 Bile acid Deficiency: Ileal Resection
Dr. Linda S. Costanzo, Ph.D.

Integrated concepts and objectives

SOM.MK.I.BPM2.3.DM.3.BCHM.1250	Outline lipid digestion and absorption and describe the significance of bile.
SOM.MK.II.BPM2.3.DM.1.PHYS.1135	Describe common causes of steatorrhea, and predict effects of steatorrhea on absorption of fat-soluble vitamins.
SOM.MK.II.BPM2.3.DM.1.PHYS.1136	Compare the absorption of fat soluble and water soluble vitamins and give examples of diseases resulting from their malabsorption.
SOM.MK.II.BPM2.3.DM.1.PHYS.1137	Describe the related roles of fluid malabsorption in the small intestine versus colon on the potential to cause diarrheal disease.
SOM.MK.I.BPM2.3.DM.3.BCHM.1244	Outline role of liver in synthesis of primary bile acids from cholesterol. Review formation of bile salts and their release into bile.
SOM.MK.I.BPM2.3.DM.3.BCHM.1243	Discuss differences between primary and secondary bile acids.
SOM.MK.I.BPM2.3.DM.3.BCHM.1251	Describe enterohepatic circulation of bile salts. Summarize functions of bile salts.
SOM.MK.I.BPM2.3.DM.3.BCHM.1249	Describe reaction catalyzed by pancreatic lipase and review importance of bile salts and colipase. Highlight its role in emulsification of dietary lipids.
SOM.MK.I.BPM2.3.DM.3.BCHM.1372	Describe metabolic basis for use of statins, ezetimibe and bile acid sequestering agents in hypercholesterolemia.

Lecture slides

This is a clinical case lecture with in-class IMCQs. You can use the main case vignette on the next slide to prepare and search for key issues related to the clinical scenario. The questions and remaining slides will be released after the lecture has been delivered. There is pre-reading material for this lecture as well, because it is a flipped classroom session. Please go through the handout on absorption before the lecture.

Case study: Ileal resection

A **39-year-old high school guidance counselor** was diagnosed with Crohn's disease (an inflammatory bowel disease) when he was a teenager.

For **20 years**, he was treated medically with **antidiarrheal** agents and strong **anti-inflammatory** drugs, including glucocorticoids.

During that time, he had two spontaneous remissions. However, after these remissions, his disease always returned "with a vengeance."

Last year, he had a **small bowel obstruction** that could not be relieved with nonsurgical approaches, and he underwent emergency surgery that removed **80% of his ileum**.

Since the surgery, he has had **diarrhea**. His stools are **oily, pale, and foul smelling**. He takes the drug **cholestyramine** to control his diarrhea. However, he continues to have steatorrhea. He also receives monthly injections of **vitamin B12**.